

Please write clearly in block capitals.

Centre number

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Candidate number

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Surname

Forename(s)

Candidate signature

I declare this is my own work.

INTERNATIONAL AS CHEMISTRY (9620)

Unit 2: Organic 1 and Physical 1

Tuesday 13 May 2025

07:00 GMT

Time allowed: 1 hour 30 minutes

Materials

For this paper you must have:

- the Periodic Table/Data Sheet, provided as an insert
- a ruler with millimetre measurements
- a scientific calculator, which you are expected to use where appropriate.

Instructions

- Use black ink or black ball-point pen.
- Fill in the boxes at the top of this page.
- Answer **all** questions.
- You must answer the questions in the spaces provided. Do **not** write outside the box around each page or on blank pages.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- All working must be shown.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 70.

For Examiner's Use	
Question	Mark
1	
2	
3	
4	
5	
6	
7	
TOTAL	



Answer **all** questions in the spaces provided.

0 1

Crude oil is a mixture of hydrocarbons.

Crude oil is separated into fractions by fractional distillation.

0 1 . 1

Give the meaning of hydrocarbon and of fraction.

[2 marks]

Hydrocarbon _____

Fraction _____

0 1 . 2

Hydrocarbons are burned to generate electricity.

During the combustion of hydrocarbons, some sulfur-containing compounds are formed.

Identify a reagent used to remove these sulfur-containing compounds.

Explain why these sulfur-containing compounds must be removed.

[2 marks]

Reagent _____

Explanation _____



The fractions that contain long chain molecules can be cracked to form short chain molecules.

0 1 . 3

Write an equation to show how one molecule of $C_{12}H_{26}$ can be cracked to form two molecules of C_4H_8 and one molecule of another hydrocarbon.

State **two** conditions used in this reaction.

[2 marks]

Equation _____

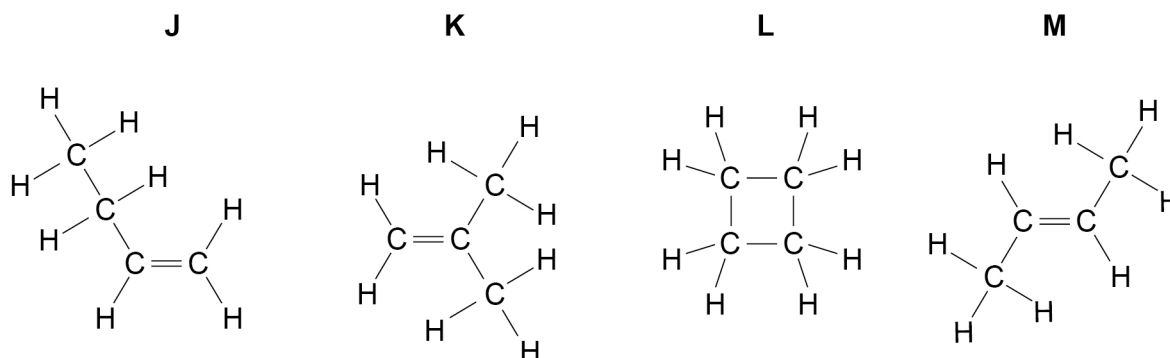
Condition 1 _____

Condition 2 _____

0 1 . 4

Figure 1 shows four isomers with the molecular formula C_4H_8

Figure 1



Identify the isomer(s) that show(s) *E-Z* isomerism.

Identify the isomer(s) that react with $Br_2(aq)$

[2 marks]

E-Z isomerism _____

Reacts with $Br_2(aq)$ _____

Question 1 continues on the next page

Turn over ►



0 1 . 5

Write an equation for the incomplete combustion of C_4H_8 to form carbon monoxide and one other product.

[1 mark]

0 1 . 6

Carbon monoxide and oxides of nitrogen can be removed from the exhaust gases of internal combustion engines in a catalytic converter.

A catalytic converter contains a thin layer of catalyst.

Write an equation to show how these gases are removed in a catalytic converter.

Explain why a thin layer of catalyst is used.

[2 marks]

Equation _____

Explanation _____

11

Turn over for the next question

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box*

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ANSWER IN THE SPACES PROVIDED**

Turn over ►



0 5

0	2
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Propane reacts with Cl_2 to produce 1-chloropropane, $\text{CH}_3\text{CH}_2\text{CH}_2\text{Cl}$

0	2	.	1
---	---	---	---

In the first stage of the mechanism, the Cl_2 molecule breaks into two $\text{Cl}^\cdot$ free radicals.

State the condition needed for this to happen.

[1 mark]

0	2	.	2
---	---	---	---

Write equations for the **two** propagation steps in the formation of $\text{CH}_3\text{CH}_2\text{CH}_2\text{Cl}$

Use structural formulas in your equations.

[2 marks]

1 _____

2 _____

0	2	.	3
---	---	---	---

Write an equation for the termination step that forms the product $\text{CH}_3(\text{CH}_2)_4\text{CH}_3$

[1 mark]



0 2 . 4 A different alkane reacts with chlorine to produce a chloroalkane with $M_r = 127.0$

The chloroalkane contains 37.8% of carbon and 6.3% of hydrogen by mass.

Calculate the molecular formula of the chloroalkane.

[4 marks]

Molecular formula _____

8

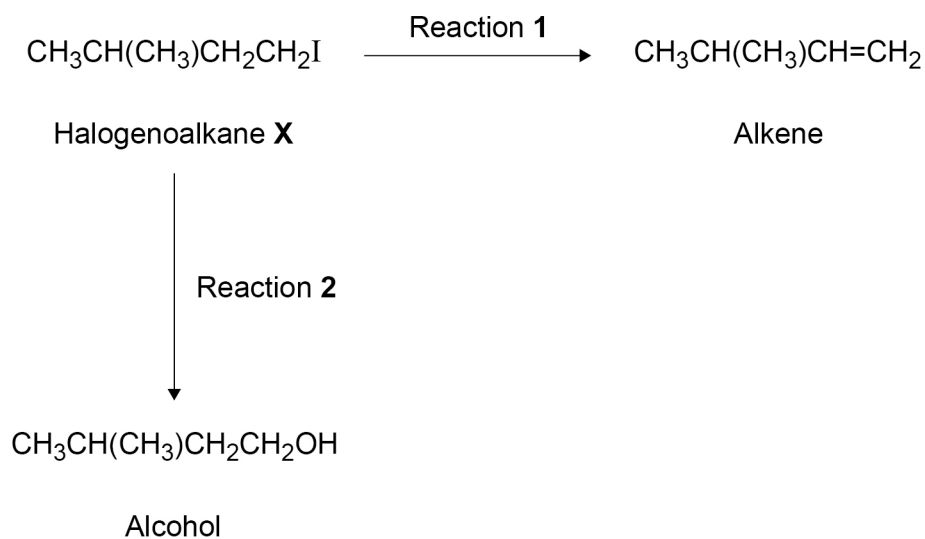
Turn over ►



0 3

Figure 2 shows two reactions of halogenoalkane **X**.

Figure 2



0 3 . 1

Use IUPAC rules to name **X**.

[1 mark]

0 3 . 2

Name the mechanism for Reaction 1.

Identify the reagent that reacts with **X** in Reaction 1.

State the condition needed for Reaction 1.

[3 marks]

Name of mechanism _____

Reagent _____

Condition _____



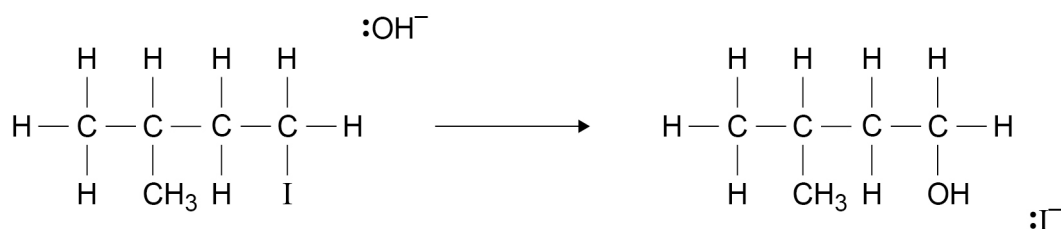
0 3 . 3 Name the mechanism for Reaction 2.

Complete the mechanism by adding curly arrows.

[2 marks]

Name of mechanism _____

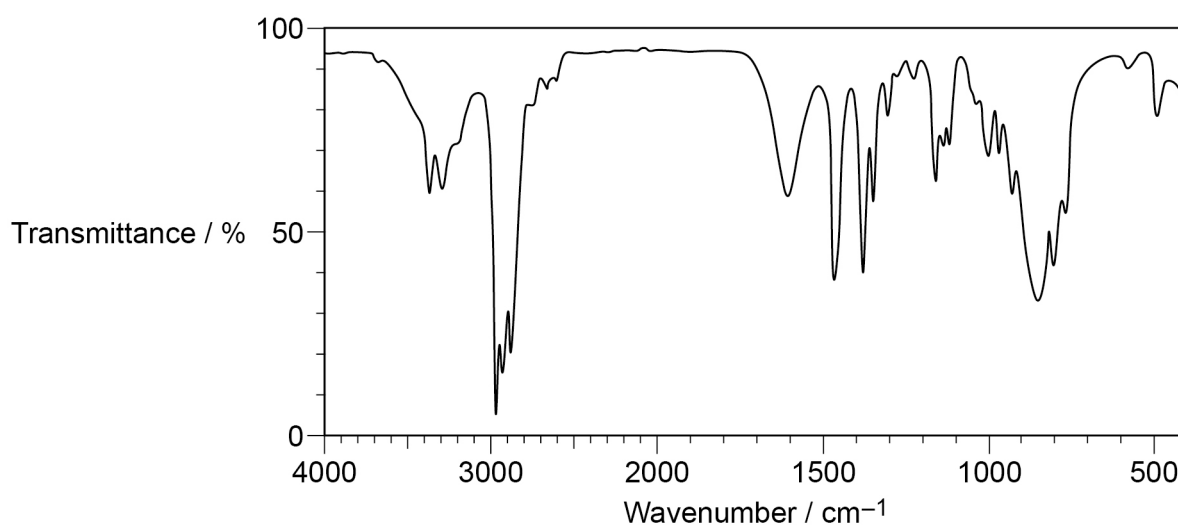
Mechanism



0 3 . 4 In a different reaction, **X** reacts to form another organic compound.

Figure 3 shows the infrared spectrum of this organic compound.

Figure 3



Identify **two** features, between 4000–2000 cm⁻¹ in the spectrum, that show that **X** reacted with NH₃ and not KCN

Use data from **Table A** on the Chemistry Data Sheet in your answer.

[2 marks]

1 _____

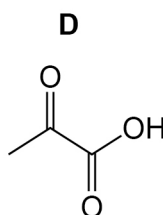
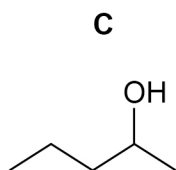
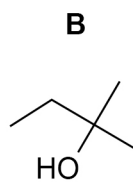
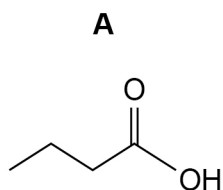
2 _____



0 4

Different organic compounds can have the same relative molecular mass.

Compounds **A**, **B**, **C** and **D** have a relative molecular mass of 88.0



These compounds can be distinguished by simple test-tube reactions.

0 4 . 1

Identify a reagent, or combination of reagents, that can be used to distinguish between separate samples of **A** and **B**.

State what is observed.

[3 marks]

Reagent(s) _____

Observation with **A** _____

Observation with **B** _____



0 4 . 2

Identify a reagent, or combination of reagents, that can be used to distinguish between separate samples of **B** and **C**.

State what is observed.

[3 marks]

Reagent(s) _____

Observation with **B** _____

Observation with **C** _____

0 4 . 3

High resolution mass spectrometry can determine relative molecular masses to four decimal places.

Table 1 shows some precise relative atomic mass values.

Table 1

Atom	Precise relative atomic mass
^1H	1.0079
^{12}C	12.0000
^{16}O	15.9994

Deduce the molecular formulas of **A** and **D**.

Use the data in **Table 1** to show how **A** and **D** can be distinguished from each other.

[3 marks]

Molecular Formula of **A** _____

Molecular Formula of **D** _____

How **A** and **D** can be distinguished from each other _____

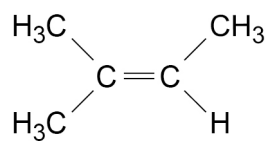
Turn over ►



0 5

This question is about alkenes.

The structure of 2-methylbut-2-ene is shown.

**0 5 . 1**

2-Methylbut-2-ene can form an addition polymer.

Draw the repeating unit of the polymer.

[1 mark]**0 5 . 2**

Explain why poly(2-methylbut-2-ene) melts at a higher temperature than 2-methylbut-2-ene.

[1 mark]



0 5 . 3 2-Methylbut-2-ene can react with HBr in an electrophilic addition mechanism.

Explain why HBr is attracted to the double bond in the alkene.

[3 marks]

0 5 . 4 When 2-methylbut-2-ene reacts with HBr, a major and minor product are formed.

Draw the structure of the major product.

Explain why there is a higher yield of the major product.

[3 marks]

Structure

Explanation _____

Turn over ►



0	5	.	5
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The major and minor products are position isomers.

Explain how infrared spectroscopy can be used to distinguish between the major and minor products from this reaction.

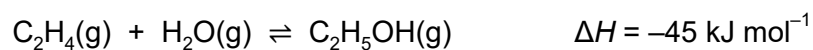
[2 marks]

10



0 6

Ethanol can be produced by the reaction of ethene with steam.



0 6 . 1

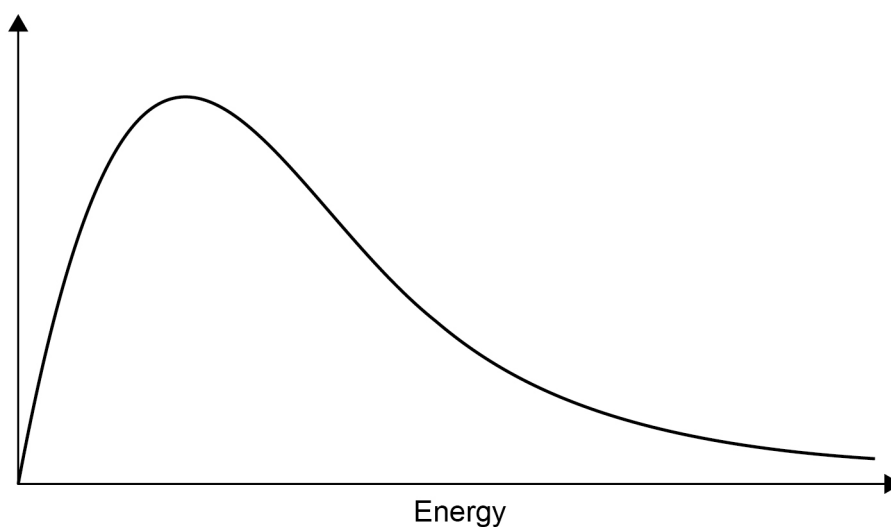
Give an expression for the equilibrium constant, K_c , for this reaction.Deduce the units of K_c

[2 marks]

 K_c

Units _____

Figure 4 shows the Maxwell–Boltzmann distribution of molecular energies for a sample of ethene at a given temperature.

Figure 4

0 6 . 2

State the label for the y-axis.

[1 mark]

0 6 . 3

Draw, on **Figure 4**, a distribution of molecular energies for the same sample of gas at a higher temperature.

[2 marks]

Turn over ►



0	6	.	4
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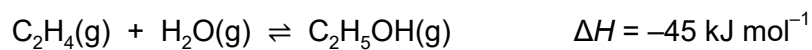
Concentrated phosphoric acid is used as a catalyst in the reaction of ethene with steam.

Explain, in general terms, how a catalyst increases the rate of a reaction.

[2 marks]

0	6	.	5
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A pressure of 6.1 MPa is used in the reaction.



Predict the effect of using a higher pressure on the rate of reaction and on the equilibrium yield of ethanol.

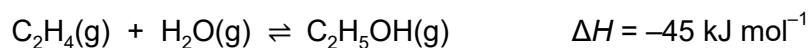
[2 marks]

Effect on rate _____

Effect on yield _____



0 6 . 6 A temperature of 300 °C is used in the reaction.



Explain why a higher temperature increases the rate of reaction.

State and explain the effect of a higher temperature on the equilibrium yield of ethanol.

Justify the use of 300 °C in the reaction.

[5 marks]

Why higher temperature increases rate _____

Effect of higher temperature on yield _____

Explanation _____

Justification _____

14

Turn over for the next question

Turn over ►



0 7

3-Methylbutan-1-ol, $\text{CH}_3\text{CH}(\text{CH}_3)\text{CH}_2\text{CH}_2\text{OH}$, is oxidised to an aldehyde.

Method

- The mass of an empty test tube is measured.
- Some 3-methylbutan-1-ol, acidified potassium dichromate(VI) and three anti-bumping granules are added to a flask.
- A condenser is connected to the flask and the flask is heated with a water bath.
- The distilled liquid products are collected in the test tube.
- The total mass of the liquid products is measured.
- The total volume of the liquid products is measured.

0 7 . 1

The total mass of the liquid products is 11.0 g
The total uncertainty in the mass measurement is ± 0.005 g

Calculate the percentage uncertainty in the mass recorded of liquid products.

[1 mark]

Percentage uncertainty _____

0 7 . 2

Why must anti-bumping granules be added to the flask?

Tick (✓) **one** box.

[1 mark]

To act as a catalyst for the reaction

☐

To prevent formation of large bubbles

☐

To prevent the reagents from boiling

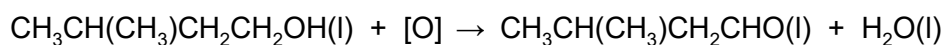
☐

07.3

State why a Bunsen burner is **not** used to heat the flask.

[1 mark]

07.4

10.0 cm³ of CH₃CH(CH₃)CH₂CH₂OH (*M*_r = 88.0) are used in the reactionDensity of CH₃CH(CH₃)CH₂CH₂OH = 0.809 g cm⁻³Density of CH₃CH(CH₃)CH₂CHO = 0.785 g cm⁻³Calculate the maximum volume of CH₃CH(CH₃)CH₂CHO that can be collected in this experiment.

Give your answer to 3 significant figures.

[3 marks]

Volume of CH₃CH(CH₃)CH₂CHO _____ cm³

Turn over ►



0 7 . 5

3-Methylbutan-1-ol ($\text{CH}_3\text{CH}(\text{CH}_3)\text{CH}_2\text{CH}_2\text{OH}$) can also react with hot concentrated sulfuric acid in an elimination reaction to form an alkene.

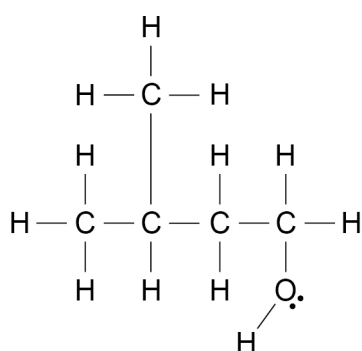
Explain why concentrated sulfuric acid is a catalyst in this reaction.

Outline the mechanism for this reaction.

[4 marks]

Explanation _____

Mechanism



END OF QUESTIONS



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[illegible]

